

Low-Resource Speaker Verification for Sinhala and Tamil

An open-source toolkit, a reproducible benchmark, and a single-corpus
empirical study

[Your Name]

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Abstract

Speaker verification (SV) systems have converged on architectures and training recipes optimised for high-resource English benchmarks; Sinhala and Tamil, spoken by over 22 million Sri Lankans, have no published SV baselines. This thesis contributes (i) a reproducible 527-speaker Sinhala/Tamil benchmark built from public read-speech corpora (OpenSLR SLR52/SLR65), with an in-the-wild extension via our group’s 280-speaker SLCeleb corpus; (ii) the first zero-shot SV baseline table for Sinhala: four modern VoxCeleb-trained architectures degrade $2\text{--}8\times$ in EER under the language shift, with badly calibrated scores throughout; (iii) a training-free adaptation study showing AS-Norm alone recovers up to 39% relative EER (best training-free system: ReDimNet-B6 + AS-Norm at 2.07%/0.90% EER si/ta), that PLDA’s value is condition-dependent, and that cross-language calibrator transfer inflates C_{llr} up to $5.7\times$ — per-language calibration is mandatory; and (iv) a four-policy fine-tuning comparison of the WavLM-base+ foundation model in which full fine-tuning (1.69%/1.41% EER over three seeds) decisively beats LoRA on this benchmark, contrary to recent PEFT findings, with the closed-set caveat analysed. The underlying toolkit extends the open-source `voxceleb_trainer` codebase with eleven feature modules and twenty-six bug fixes, and all trial lists, configurations, and result artifacts are released for downstream Sri Lankan SV research.

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Chapter 1

Introduction

1.1 Motivation

Speaker verification (SV) — the task of deciding whether two audio recordings come from the same speaker — is a foundational biometric technology underpinning voice authentication, forensic audio analysis, and conversational-AI personalisation. The dominant publicly available benchmarks (VoxCeleb1 [11], VoxCeleb2, NIST SRE series) are constructed almost exclusively from English speech. As a result, the state-of-the-art has converged on architectures and training recipes that, while highly effective on English, are not guaranteed to generalise to low-resource languages.

Sri Lanka presents a particularly underserved case. The country’s two primary languages — Sinhala (an Indo-Aryan language with roughly 17 million speakers) and Tamil (a Dravidian language with roughly 5 million Sri Lankan speakers) — have, prior to this work and our group’s SLCeleb corpus [17], no publicly catalogued SV benchmarks and no published SV baselines. Existing multilingual self-supervised speech encoders cover them only as part of broader low-resource batches [2, 1], with no targeted SV evaluation. The Sri Lankan deployment context further compounds the difficulty: bilingual and code-switched speech is normal in everyday usage, telephony channels dominate practical deployment, and labelled speakers are expensive to collect.

1.2 Problem statement

This thesis addresses the question of how to construct a speaker verification system that performs reliably on Sinhala, Tamil, and code-switched audio when only a small labelled corpus is available. We frame the problem along three dimensions:

1. **Architectural choice.** Modern SV architectures (ECAPA-TDNN [4], ResNet-SE variants, self-supervised front-ends) were validated on English VoxCeleb scale. Whether their inductive biases hold for low-resource Sinhala/Tamil is an open empirical question.
2. **Cross-lingual transfer.** When a target-language corpus is small ($\leq 10\,000$ utterances), training from scratch overfits. Cross-lingual fine-tuning from an English-pretrained checkpoint is the natural alternative; the right freezing, learning-rate,

and regularisation policy for SV fine-tuning specifically remains under-studied.

3. **Calibration and operating point.** Per-language decision thresholds, score normalisation (AS-Norm [10]), generative scoring backends (PLDA), and language-aware regularisation (domain adversarial training [5]) are standard tools for cross-lingual SV in the literature, but their interactions on Sinhala/Tamil data are not characterised.

1.3 Research questions

We address the following research questions:

- RQ1** How well do released, English/VoxCeleb-trained SV models transfer *zero-shot* to Sinhala and Tamil, and how large is the degradation relative to their published English benchmarks?
- RQ2** Which fine-tuning policy — frozen-encoder probing, parameter-efficient adaptation (LoRA), full fine-tuning, or layer-wise LR decay — best adapts a pretrained speech foundation model to Sinhala/Tamil per trainable parameter?
- RQ3** Does language-adversarial training (gradient reversal / DANN [5]) improve cross-lingual verification of bilingual speakers, and by how much?
- RQ4** At eval time, do score normalisation (AS-Norm) and a generative scoring backend (PLDA) compose constructively, or do they redundantly model the same residual variance?
- RQ5** How large is the gap between monolingual (Sinhala-only, Tamil-only) and cross-lingual EER for a single trained model, and can it be closed via threshold calibration alone or does it require architectural intervention?

1.4 Contributions

The contributions of this thesis are:

1. **A reproducible Sinhala/Tamil SV benchmark.** Benchmark v0 covers 527 speakers (478 Sinhala from OpenSLR SLR52, 49 Tamil from OpenSLR SLR65) with seed-reproducible monolingual trial lists (12,000 trials per language, gender-controlled impostors where metadata permits), produced by a released deterministic pipeline; benchmark v1 extends the evaluation to SLCeleb [17], our group’s 280-speaker in-the-wild corpus with multi-session and bilingual speakers.
2. **A comprehensive empirical study** of cross-lingual transfer for SV under low-resource constraints, in three suites: the first zero-shot baseline table for Sinhala (four modern architectures); a training-free adaptation ablation (AS-Norm, PLDA, per-language calibration) that quantifies how much of the cross-lingual degradation is a fixable score-domain problem; and a four-policy fine-tuning comparison (frozen / LoRA / full / layer-wise LR decay) of a WavLM foundation model, replicated over three seeds.

3. **Per-language and pooled evaluation infrastructure** that supports principled comparison of Sinhala-only, Tamil-only, and cross-lingual trial sets in a single trainer invocation, with paper-grade statistical reporting (mean $\pm$ std over three random seeds).
4. **A practical open-source toolkit** extending the `voxceleb_trainer` codebase [12] with eleven new features (FEATURE-001..FEATURE-011) and twenty-six bug fixes (BUGFIX-001..BUGFIX-026). Each feature ships with a design document and is selectable via a single YAML key, enabling independent ablation.

1.5 Thesis structure

The remainder of the thesis is organised as follows. Chapter 2 reviews prior work on speaker verification architectures, cross-lingual transfer, score normalisation, and language-aware modelling. Chapter 3 describes the corpus, architectures, training recipe, scoring backends, and evaluation protocol. Chapter 4 reports the headline and ablation results. Chapter 5 analyses the findings, addresses the research questions of Section 1.3, and discusses limitations. Chapter 6 summarises the contributions and outlines future work, including the SLCeleb (v1) open-set benchmark, the bilingual cross-lingual study, and self-supervised pretraining on unlabelled Sri Lankan audio.

Chapter 2

Related Work

2.1 Speaker verification architectures

The dominant architectures for modern speaker verification fall into three families. The first is the residual convolutional family, typified by ResNet-SE34V2 with squeeze-and-excitation blocks and self-attentive pooling. The second is the time-delay neural network (TDNN) family, dominated since 2020 by ECAPA-TDNN [4]. ECAPA-TDNN introduced three key ingredients: (i) Res2Net-style multi-scale convolutions with hierarchical accumulation, (ii) squeeze-and-excite channel re-weighting at each block, and (iii) multi-layer feature aggregation followed by channel-attentive statistical pooling. ECAPA-TDNN consistently reports 0.8–1.5 % EER on VoxCeleb1-O and has become the de-facto reference architecture in subsequent SV papers. The third family is self-supervised, replacing the mel-front-end with a frozen pretrained transformer encoder such as WavLM [2], XLS-R, or mHuBERT-147; these models typically outperform mel-based architectures on low-resource language data by 30–50 % relative EER, at the cost of significant parameter count (95M–300M).

2.2 Loss functions for speaker discrimination

Speaker-verification training has progressively replaced cross-entropy softmax with margin-based variants. AM-Softmax and Additive Angular Margin Softmax (AAM-Softmax) [3] reshape the classification objective to produce more discriminative embeddings on the hypersphere by adding a margin to the cosine similarity between the embedding and the target-class weight vector. The margin-based losses became standard in SV largely because they produce embeddings that score well under cosine distance at eval time — a closer match to how SV systems are actually used than the intra-class compactness produced by plain softmax.

Auxiliary use of language information has also proven productive, either during training via multi-task language-ID heads, or at decision time via language-aware scoring and calibration [18]. Adversarial counterparts — Domain-Adversarial Neural Networks [5] attached to a language or channel classifier through a gradient-reversal layer — pursue the opposite goal: forcing the embedding to be *uninformative* about the domain. Both approaches appear in the SV literature; their relative effectiveness depends on whether language is a useful signal (multi-task wins) or a confound (adversarial wins).

2.3 Cross-lingual and low-resource transfer

When a target-language SV corpus is small ($\leq 10\text{k}$ utterances), training a deep architecture from scratch overfits catastrophically. The standard remedy is cross-lingual fine-tuning from a checkpoint pretrained on a high-resource corpus (typically VoxCeleb1 or VoxCeleb2). Three knobs control how aggressively the target language overwrites the pretrained representation:

1. **Selective freezing.** Freezing the front-end (SincConv or mel-filterbank parameters) preserves the acoustic-feature extractor learned on the source language. For very small corpora, freezing the entire encoder and training only the projection head is the safest option, and already captures most of the self-supervised representation’s value [2, 15].
2. **Learning-rate scaling.** A standard multiplier of $0.1\times$ to $0.01\times$ the from-scratch LR avoids destabilising the prior.
3. **Layer-wise learning-rate decay (LLRD).** A decay factor $\gamma \in [0.8, 0.95]$ applied per layer from the encoder top to the input front-end, so that deeper layers carry transferable acoustic priors while shallower (closer-to-output) layers adapt more aggressively. LLRD is universal in the BERT and WavLM fine-tuning literature; its application to SV fine-tuning is less systematically reported, motivating one of this thesis’s ablations.

2.4 Score normalisation and calibration

Two eval-time mechanisms address the gap between training-time and deployment-time score distributions. Adaptive Symmetric Score Normalisation (AS-Norm) [10] computes per-trial-side cohort statistics from a fixed set of impostor embeddings and normalises trial scores against those statistics. It is universal in the NIST SRE pipeline and typically yields 5–20% relative MinDCF reduction at zero training cost.

Probabilistic Linear Discriminant Analysis (PLDA) [8, 16] models the embedding distribution generatively and replaces cosine scoring with a likelihood ratio test under “same speaker” vs “different speaker” hypotheses. PLDA was the dominant SV backend in the i-vector era and remains useful for short-utterance and channel-mismatched trials, although on modern discriminatively-trained embeddings the marginal gain over cosine is smaller. The two-covariance simplification of PLDA [16] provides nearly identical EER to the full formulation at half the computational complexity.

Score normalisation and PLDA compose: AS-Norm normalises against cohort statistics regardless of whether the underlying score is cosine or PLDA. The NIST SRE pipeline order is PLDA $\rightarrow$ AS-Norm.

2.5 Multilingual self-supervised speech representation

Self-supervised models trained on large multilingual audio corpora (wav2vec 2.0 [1], WavLM [2], XLS-R, mHuBERT-147 [21]) provide speech-aware embeddings without

language-specific labelling. mHuBERT-147 is particularly relevant to the Sri Lankan setting: it deliberately upweights low-resource languages including Sinhala and Tamil in its pretraining corpus.

There are three ways to use such models for low-resource SV: (i) load an off-the-shelf checkpoint, freeze it, and train only a pooling head and projection on the target-language labelled data pretraining (CPT) on unlabelled target-language audio before SV-fine-tuning, which adds 5–15 % relative EER on top of (i) when sufficient unlabelled audio (≥ 500 h) is available [13]; (iii) pretrain from scratch on target-language-only audio, which requires thousands of hours and rarely outperforms (ii). This thesis evaluates option (i) as the SSL baseline; option (ii) is documented as future work.

2.6 Positioning of this work

The contributions outlined in Section 1.4 sit at the intersection of three otherwise separately-developed lines of work:

- **Architectures and losses** for English-scale SV (§ 2.1, § 2.2) are applied to low-resource Sinhala/Tamil with no targeted benchmark in the literature.
- **Cross-lingual transfer techniques** (§ 2.3) are widely studied for ASR and translation but less so for SV; their interaction with SV-specific losses (AAM-Softmax) and SV-specific eval backends (PLDA, AS-Norm) has not been systematically ablated.
- **Multilingual SSL encoders** (§ 2.5) cover Sinhala and Tamil only as part of broader low-resource batches, with no published SV evaluation on either language.

To the best of our knowledge, this is the first study to provide a reproducible SV benchmark on Sinhala and Tamil with full ablation isolation of the cross-lingual transfer levers. The benchmark, the corpus preparation pipeline, and the open-source toolkit (§ 1.4) are released to support future work in this direction.

Chapter 3

Methods

3.1 Corpus and benchmark construction

We evaluate on a two-tier benchmark. **Benchmark v0**, used for all experiments in this thesis, is built from two public read-speech corpora: OpenSLR SLR52 [9] contributes 478 Sinhala speakers ($\approx 185k$ crowdsourced utterances, 16 kHz FLAC, minimum 10 utterances per speaker), and OpenSLR SLR65 [6] contributes 49 Tamil speakers (4,286 studio utterances, 48 kHz resampled at load). Speaker identities come from the corpora’s published metadata; gender labels exist only for SLR65. A deterministic ingestion script (`tools/ingest_openslr.py`) lays the corpus out as `<lang>/<speaker>/<utterance>` symlinks and emits a manifest and speaker-metadata table. **Benchmark v1**, pending data staging, replaces the evaluation side with SLCeleb [17], our group’s 280-speaker in-the-wild Sinhala/Tamil celebrity corpus, which adds multi-genre audio, multi-session speakers, and bilingual speakers for true cross-lingual trials.

3.2 Trial construction

Utterances are split 80/20 per speaker into train and test partitions (seed 42, `tools/sl_dataprep.py`). From the test partition we sample, per language, 3,000 same-speaker target pairs and 9,000 same-language impostor pairs (1:3 ratio), with unordered-pair deduplication and attempt-bounded sampling. Impostor pairs are same-gender whenever both speakers’ genders are known; this holds for all Tamil trials but cannot be enforced for Sinhala (SLR52 has no gender metadata). Cross-session targets are enforced when session keys are derivable; the OpenSLR layouts carry no session metadata, so v0 targets are not session-controlled. Consequently v0 absolute error rates are optimistic and are valid for *relative* comparison between systems, not for comparison against in-the-wild benchmarks; we state this caveat wherever v0 numbers are reported. The cross-lingual trial list is empty at v0 (the OpenSLR corpora contain no bilingual speakers) and becomes meaningful at v1.

3.3 Metrics

We report equal error rate computed as the mean of the false-positive and false-negative rates at the crossing point (EER_{avg}), and minimum detection cost (minDCF) at two operating points, $p_{\text{target}} \in \{0.01, 0.05\}$ with $C_{\text{miss}} = C_{\text{fa}} = 1$, following NIST and VoxCeleb practice. Calibration quality is measured with C_{llr} and actual DCF after logistic calibration. Unlabeled trials are treated as a hard error by the evaluation harness.

3.4 Experiment suite A: zero-shot baselines

Four released, VoxCeleb-trained checkpoints are evaluated with no adaptation of any kind: ECAPA-TDNN [4] as distributed by SpeechBrain [14] ($\approx 20\text{M}$ parameters), and ReDimNet-B1/B2/B6 [20] (2.2M/4.7M/15M). Scoring is cosine similarity on L2-normalised full-utterance embeddings, with no score normalisation and no calibration; this table is the raw-transfer reference that all adaptation levers are measured against.

3.5 Experiment suite B: training-free backend adaptation

On top of cached suite-A embeddings we ablate three eval-time levers, none of which updates the encoder: (i) **AS-Norm** [10]: adaptive symmetric score normalisation with a 527-utterance in-language cohort (one utterance per training-pool speaker), top- $K=300$; (ii) **PLDA** [16, 8]: a two-covariance model ($d=150$) trained on 10,480 utterances (≤ 20 per speaker) from the training partition, preceded by centering, LDA, and length normalisation; (iii) **language-aware logistic calibration**: an affine score-to-LLR map fitted per language on half the trials and evaluated on the other half, plus the cross-language transplant of each calibrator to quantify the cross-lingual score shift [18]. The grid crosses $\{\text{cosine}, \text{PLDA}\} \times \{\text{raw}, \text{AS-Norm}\}$ for three encoders, answering RQ4 directly. The PLDA fit necessarily sees the trial speakers (utterance-disjoint, standard for domain-adaptation PLDA); we flag this closed-set property explicitly.

3.6 Experiment suite C: parameter-efficient fine-tuning

Suite C fine-tunes WavLM-base+ [2] (94.4M parameters, English-pretrained) for Sinhala/Tamil speaker verification. The 13 hidden states are combined by a learnable softmax-weighted sum, pooled by attentive statistics pooling (128-d bottleneck), and projected to a 192-d embedding trained with AAM-Softmax [3] (margin 0.2, scale 30) over the 527 training speakers. The CNN feature extractor stays frozen in all arms. Four fine-tuning policies are compared:

frozen encoder frozen; only the pooling/embedding head and layer weights train ($\approx 0.5\%$ of parameters);

LoRA rank-8 adapters ($\alpha=16$) [7] on the query/key/value/output projections of every attention block, plus the head (1.08M trainable, 1.1%);

full all encoder parameters unfrozen (encoder LR 10^{-5} , head LR 10^{-3});

LLRD full fine-tuning with layer-wise learning-rate decay ($\gamma=0.9$ per layer from the top).

Training uses the v0 train split capped at 60 utterances per speaker (31,049 utterances), random 2.0s crops, batch 64, AdamW (weight decay 2×10^{-5}), exponential LR decay ($\gamma=0.95$), mixed precision, gradient clipping at 5.0, and 15 epochs. Model selection uses per-epoch pooled EER on stratified 2,000-pair subsamples; the selected checkpoint is finally evaluated on the full 12,000-pair lists, making suite-C numbers directly comparable to suites A and B. Because the evaluation speakers are the training speakers (utterance-disjoint split), suite C measures *representation adaptation* rather than open-set generalisation; the v1 rerun provides the open-set condition. Finally, the suite-B backend stack is re-applied to the fine-tuned embeddings (suite B $\times$ C composition) to test whether backend adaptation retains value once the encoder is in-domain [19].

3.7 Reproducibility

Trial lists, cohort lists, and PLDA training lists are reproducible from seed 42. The LoRA and full arms of suite C are replicated with three seeds (42/43/44) and reported as mean $\pm$ std; the frozen and LLRD arms use seed 42 only, as both sit far from the decision boundary between policies. All runs, configurations, per-epoch histories, and raw result JSONs are archived with the research logs listed in the chapter header.

Chapter 4

Results

4.1 Zero-shot baselines (suite A)

Table 4.1 reports, to our knowledge, the first published speaker-verification results for Sinhala. All four VoxCeleb-trained checkpoints transfer at usable but clearly degraded accuracy: relative to each model’s published VoxCeleb1-O EER, Sinhala EER inflates by $4.9\text{--}8.0\times$ and Tamil by $2.3\text{--}4.0\times$ — Tamil within the $2\text{--}4\times$ band the CN-Celeb cross-domain literature reports, Sinhala well above it, even though v0 is clean read speech. The VoxCeleb ranking is mostly preserved (ReDimNet-B6 best, ECAPA worst) with one inversion: ReDimNet-B1 beats the nominally stronger B2 on both languages, and with 2.2M parameters also beats the $\approx 20\text{M}$ -parameter ECAPA — parameter count does not monotonically buy language robustness. Across all systems $\text{minDCF}(10^{-2})$ is poor ($0.24\text{--}0.48$) while EERs look usable: the signature of badly scaled scores in a mismatched domain, which motivates suite B.

4.2 Training-free backend adaptation (suite B)

Table 4.2 crosses the scoring backend (cosine vs PLDA) with AS-Norm for three encoders. Three findings:

1. **AS-Norm delivers 12–39% relative EER reduction** (Sinhala: $33/33/23\%$ for ECAPA/B1/B6; Tamil: $30/12/39\%$), bracketing the $20\text{--}30\%$ band reported by [10], and improves $\text{minDCF}(10^{-2})$ in five of the six model–language cells (best: B6 Sinhala $0.238 \rightarrow 0.167$; the exception is B1 Tamil, $0.406 \rightarrow 0.433$). It is the most cost-effective lever in this study: no training, one cohort file.
2. **PLDA hurts Sinhala but helps Tamil.** PLDA increases Sinhala EER for every encoder (B6: $2.71 \rightarrow 3.84$) while reducing Tamil EER for two of the three encoders (ECAPA: $3.21 \rightarrow 2.46$; B6: $1.48 \rightarrow 1.11$; B1 worsens slightly, $1.67 \rightarrow 1.77$). Our working hypothesis is a benchmark artifact: the PLDA fit is dominated by the 478 Sinhala speakers of near-single-session crowdsourced audio, so its within-class covariance underestimates true channel variability, whereas the studio-recorded Tamil fits the model. The v1 multi-session rerun is the decisive test before dismissing the “PLDA wins under shift” claim of [19].

Table 4.1: Zero-shot transfer of VoxCeleb-trained checkpoints to benchmark v0. EER_{avg} % and minDCF at $p_t=0.01/0.05$; 12,000 trials per language. $\times\text{Vox1-O}$ = SL EER divided by the checkpoint’s published VoxCeleb1-O EER.

Model	Sinhala			Tamil		
	EER	minDCF	$\times\text{Vox1-O}$	EER	minDCF	$\times\text{Vox1-O}$
ECAPA-TDNN (SpeechBrain)	4.78	0.471/0.291	6.0	3.21	0.482/0.207	4.0
ReDimNet-B1	3.57	0.306/0.209	4.9	1.67	0.406/0.162	2.3
ReDimNet-B2	4.17	0.422/0.244	8.0	1.97	0.454/0.170	3.8
ReDimNet-B6	2.71	0.238/0.153	7.3	1.48	0.422/0.160	4.0

Table 4.2: Suite B ablation: EER % (minDCF at 10^{-2}). Cosine rows reproduce Table 4.1 exactly (sanity check). Bold = best per encoder and language.

Encoder	Backend	Sinhala	Tamil
ECAPA-TDNN	cosine	4.78 (0.471)	3.21 (0.482)
	cosine+AS-Norm	3.19 (0.330)	2.24 (0.480)
	PLDA	6.00 (0.598)	2.46 (0.495)
	PLDA+AS-Norm	5.80 (0.482)	2.06 (0.374)
ReDimNet-B1	cosine	3.57 (0.306)	1.67 (0.406)
	cosine+AS-Norm	2.38 (0.245)	1.47 (0.433)
	PLDA	5.34 (0.452)	1.77 (0.419)
	PLDA+AS-Norm	5.23 (0.382)	1.49 (0.297)
ReDimNet-B6	cosine	2.71 (0.238)	1.48 (0.422)
	cosine+AS-Norm	2.07 (0.167)	0.90 (0.376)
	PLDA	3.84 (0.293)	1.11 (0.340)
	PLDA+AS-Norm	3.56 (0.254)	1.16 (0.314)

3. **RQ4 (composition):** at v0, AS-Norm dominates PLDA and the composed stack does not beat cosine+AS-Norm on Sinhala, though it yields the best Tamil minDCF. The answer is condition-dependent; both levers stay in the v1 protocol.

The best training-free v0 system is **ReDimNet-B6 + AS-Norm: 2.07% (si), 0.90% (ta)**.

4.2.1 Cross-lingual score shift and calibration

Table 4.3 fits an affine logistic calibrator per language and then transplants it across languages (ReDimNet-B6). Self-calibration achieves good C_{lr} (0.066–0.171), but the cross-language transplant inflates C_{lr} by up to $5.7\times$ (Sinhala scored with a Tamil-fit calibrator, cosine + AS-Norm). The cross-lingual score shift is therefore large and real, extending [18] to Sinhala/Tamil, and **per-language calibration is mandatory for deployment**. A publishable nuance: PLDA scores are the most *transportable* across languages (penalty only $1.2\text{--}2.7\times$) even where PLDA’s EER is worse — the generative backend produces more language-homogeneous score scales than cosine.

Table 4.3: Language-aware calibration, ReDimNet-B6: C_{lr} with a self-language-fit calibrator vs a calibrator transplanted from the other language.

Backend	Sinhala		Tamil	
	self	ta-fit	self	si-fit
cosine	0.129	0.489 (3.8 $\times$)	0.095	0.149 (1.6 $\times$)
cosine+AS-Norm	0.099	0.565 (5.7 $\times$)	0.071	0.167 (2.4 $\times$)
PLDA	0.171	0.390 (2.3 $\times$)	0.073	0.084 (1.2 $\times$)
PLDA+AS-Norm	0.163	0.442 (2.7 $\times$)	0.066	0.094 (1.4 $\times$)

Table 4.4: Suite C: fine-tuning policy comparison on benchmark v0 (EER_{avg} %, full 12,000-pair lists, cosine scoring). Seed-42 column for all arms; mean $\pm$ std over seeds 42/43/44 for LoRA and full. Zero-shot reference: ReDimNet-B6 2.71 (si) / 1.48 (ta).

Arm	Trainable	seed 42		mean $\pm$ std (3 seeds)	
		si	ta	si	ta
frozen	0.5%	3.74	3.13	—	—
LoRA	1.1%	3.17	2.44	3.73 $\pm$ 0.49	3.11 $\pm$ 0.61
LLRD	95.6%	2.40	1.81	—	—
full	95.6%	1.63	1.30	1.69$\pm$0.06	1.41$\pm$0.10

4.3 Parameter-efficient fine-tuning (suite C)

Table 4.4 compares the four fine-tuning policies. Accuracy is monotone in encoder freedom: frozen $\prec$ LoRA $\prec$ LLRD $\prec$ full. Full fine-tuning of WavLM-base+ on 31k in-language utterances is the best system in this thesis — **1.69 $\pm$ 0.06% (si) / 1.41 $\pm$ 0.10% (ta)** over three seeds — beating the best zero-shot supervised model by 38% relative on Sinhala (2.71% $\rightarrow$ 1.69%) and matching it within noise on Tamil.

Two observations qualify the PEFT narrative:

1. **The “PEFT $\geq$ full fine-tuning under shift” prediction [15] does not reproduce here:** full fine-tuning beats LoRA at every seed, with a 2.2 $\times$ mean EER gap (si 1.69 vs 3.73). A plausible explanation is the closed-set v0 protocol (Section 3.6): full fine-tuning’s usual penalty — overfitting the adaptation speakers at the cost of unseen speakers — is not measured when the evaluation speakers *are* the training speakers. The open-set v1 rerun arbitrates. Secondary factors: LoRA $r=8$ may be undersized and its head LR untuned.
2. **LoRA is high-variance across seeds** (± 0.49 –0.61 EER, vs ± 0.06 –0.10 for full): single-seed PEFT-vs-full comparisons on corpora of this size are unsound, and the seed-42-only column would have flattered LoRA with its best draw.

Even the frozen probe (0.5% of parameters) lands within ≈ 1 EER point of zero-shot ReDimNet-B6 on Sinhala — the pretrained WavLM features already carry most of the speaker signal, and everything beyond frozen is adaptation gain.

Table 4.5: Suite B backends applied to fine-tuned WavLM embeddings (full arm, seed 42): EER % (minDCF at 10^{-2}).

Backend	Sinhala	Tamil
cosine	1.63 (0.180)	1.30 (0.175)
cosine+AS-Norm	1.71 (0.144)	1.13 (0.149)
PLDA	2.46 (0.258)	1.57 (0.234)
PLDA+AS-Norm	2.37 (0.197)	1.43 (0.201)

4.4 Composing backend adaptation with fine-tuning (suite B×C)

Re-applying the suite-B stack to the fine-tuned (full, seed 42) embeddings tests whether backend adaptation survives encoder adaptation (Table 4.5). AS-Norm’s EER gain saturates (si -0.07 absolute; ta still $+13\%$ relative) but it keeps improving the operating-point metrics — minDCF(10^{-2}) si $0.180 \rightarrow 0.144$, ta $0.175 \rightarrow 0.149$, and C_{llr} down $\approx 15\%$ on both — so it remains worth deploying even with an in-domain encoder. PLDA, by contrast, is now clearly redundant-to-harmful on both languages, exactly the in-domain prediction of [19]: fine-tuning moved the embedding distribution in-domain, and the v0 PLDA story is coherent before and after. The best overall v0 operating system is **fine-tuned WavLM (full) + AS-Norm**: si $1.71\%/0.144$, ta $1.13\%/0.149$ (or plain cosine when EER is the sole criterion).

4.5 Per-language analysis

Tamil EER is uniformly lower than Sinhala (studio recordings, 49 speakers, gender-controlled impostors), yet Tamil minDCF(10^{-2}) is uniformly *worse* for the ReDimNet family (e.g. B6: 0.238 si vs 0.422 ta). With only 49 speakers the impostor pool is small, and a few hard same-genre impostors dominate the low-false-alarm operating region. Two methodological consequences: EER alone overstates how “solved” the Tamil condition is, and both metrics must be read against the corpus asymmetry (478 crowdsourced phone-quality Sinhala speakers vs 49 studio Tamil speakers) rather than interpreted as a property of the languages themselves. Cross-lingual (enrol-in-one-language, test-in-the-other) trials require bilingual speakers and therefore await benchmark v1.

4.6 Statistical reporting

Multi-seed replication (42/43/44) covers the two arms whose ordering constitutes the headline claim (LoRA vs full); their separation (at least 1.1 EER points at every matched seed, and non-overlapping seed ranges on both languages) makes the verdict robust without interval estimation. The frozen and LLRD arms, and all training-free results, are deterministic given the seed-42 lists. For the journal version, bootstrap confidence intervals over trial pairs (1,000 resamples, 2.5/97.5 quantiles) remain planned once the v1 benchmark is in place.

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